Assignment 5: PH4107
Kramers Kronig Relations

Marks: 25
Due date: *Oct 14, 2026*

1. From your concepts of reflectivity and its dependencies, argue objectively the necessity of having the Kramers Kronig relations. What would the experimental measurement of reflectivity/absorptivity then enable? [Hint: you may consult the text book 'Kramers-Kronig Relations in Optical Materials Research' by Dr. Valerio Lucarini, et al.] [3+2]

2. Use the Kramers-Kronig relation to calculate the real part of $\epsilon(\omega)$, given the imaginary part of $\epsilon(\omega)$ for positive ω as
   (a) $\epsilon_2(\omega) = \lambda\left[\theta(\omega-\omega_1) - \theta(\omega-\omega_2)\right],\ \omega_2 > \omega_1 > 0.$
   (b) $\epsilon_2(\omega) = \dfrac{\lambda\gamma\omega}{\left(\omega_0^2-\omega^2\right)^2 + \gamma^2\omega^2}$
   In each case sketch the behaviour of Im $\epsilon(\omega)$ and the result for Re $\epsilon(\omega)$ functions of ω [4+6]

3. Discuss the extension of the Kramers-Kronig relations for a medium with a static electrical conductivity σ. Show that the equation for the real part of $\epsilon(\omega)$ is unchanged but the equation for the imaginary part is changed into

$$Im\ \epsilon(\omega) = \frac{\sigma}{\omega} - \frac{2\omega}{\pi} P \int_0^\infty \frac{[Re\ \epsilon(\omega') - 1]}{\omega'^2 - \omega^2} d\omega'$$

   [Hint: Consider $\epsilon(\omega) - i\sigma/\omega$ as analytic for Im $\omega \geq 0$] [5]

4. When we discussed the Drude's model for metals, we found that conductivity(σ) is a complex quantity and it's a response function. Discuss intuitively why conductivity is a response function. Derive the real part of the conductivity from the imaginary part and vice-versa using Kramers-Kronig relations. [1+2+2]